

CASE STUDY: OPTIMIZING ESTIMATION ACCURACY & COST CONTROL

Bridging the Gap Between Theoretical Quoting and Shop-Floor Reality

Executive Summary

Project margins in the Railing Division were at risk due to a disconnect between theoretical estimation formulas and actual production efforts. By implementing a system-level optimization driven by **Time Studies, Gemba Walks, and Database Sanitization**, I improved pricing accuracy, reduced manual quoting time, and eliminated \$900K+ in systemic data and inventory inaccuracies.

1. CONTEXT & CHALLENGE

Estimation is the heartbeat of project profitability. In the Railing Division, quoting was primarily based on historical assumptions and theoretical calculations, leading to a significant variance between "Estimated" and "Actual" production costs.

- **The Problem:** The estimation system was decoupled from manufacturing reality.
- **Financial Risk:** Inaccurate quotations created a high risk of margin erosion and inefficient capital allocation.
- **Systemic Gaps:**
 - Estimation values relied on outdated assumptions rather than live production data.
 - The product database was cluttered with obsolete items, causing selection errors.
 - A mismatch existed between ERP inventory records and physical shop-floor stock.

2. MANAGEMENT STRATEGY (GROUND-LEVEL VALIDATION)

I approached this as a **System-Level Disconnect** between the Commercial and Operations departments.

- **Time & Motion Studies:** Conducted rigorous on-site timing of production processes to replace "assumptions" with "validated data."
- **Data Sanitization:** Performed a comprehensive audit of the estimation database and physical inventory to align digital records with reality.
- **Gap Analysis:** Compared "Estimated vs. Actual" production times to quantify deviations and identify where labor costs were being underestimated.

3. THE SOLUTION: SYSTEM OPTIMIZATION & TOOL DEVELOPMENT

I implemented a three-pillar improvement strategy to lock in cost control:

1. **Validated Estimation Sheets:** Updated all quoting templates with real-world data, incorporating realistic allowances for setup, handling, and process variability.
2. **Database & Inventory Cleanup:** * Purged obsolete products from the system to prevent quoting of non-standard items.

- Reconciled physical stock with system records to ensure material availability was accurately reflected.
3. **Digital Tool Creation:** Developed a suite of **Excel-based Estimation Tools** for:
- **Unit-Level Quoting:** Rapid estimation for standard products.
 - **Project-Level Costing:** Comprehensive financial modeling for large-scale contracts.

4. QUANTIFIED IMPACT (ROI)

Metric	Baseline (Pre-Intervention)	Post-Intervention Result
Estimation Basis	Theoretical Assumptions	Validated Production Data
Quoted Margin Risk	High (Due to Underestimation)	Minimized / Protected
Turnaround Time	Manual / Inconsistent	Rapid / Tool-Based
Inventory Alignment	Disconnected	System-Physical Sync

5. LEADERSHIP TAKEAWAYS

- **Commercial Integrity:** Accurate engineering data is the only foundation for reliable commercial pricing.
- **System Cleanup is Strategy:** A database is only useful if it is clean. Removing "noise" (obsolete data) accelerates decision-making.
- **Sustainability through Tools:** Moving from individual "expert knowledge" to "systematized Excel tools" ensures the business can scale without losing accuracy.

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